

Solution Architecture

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

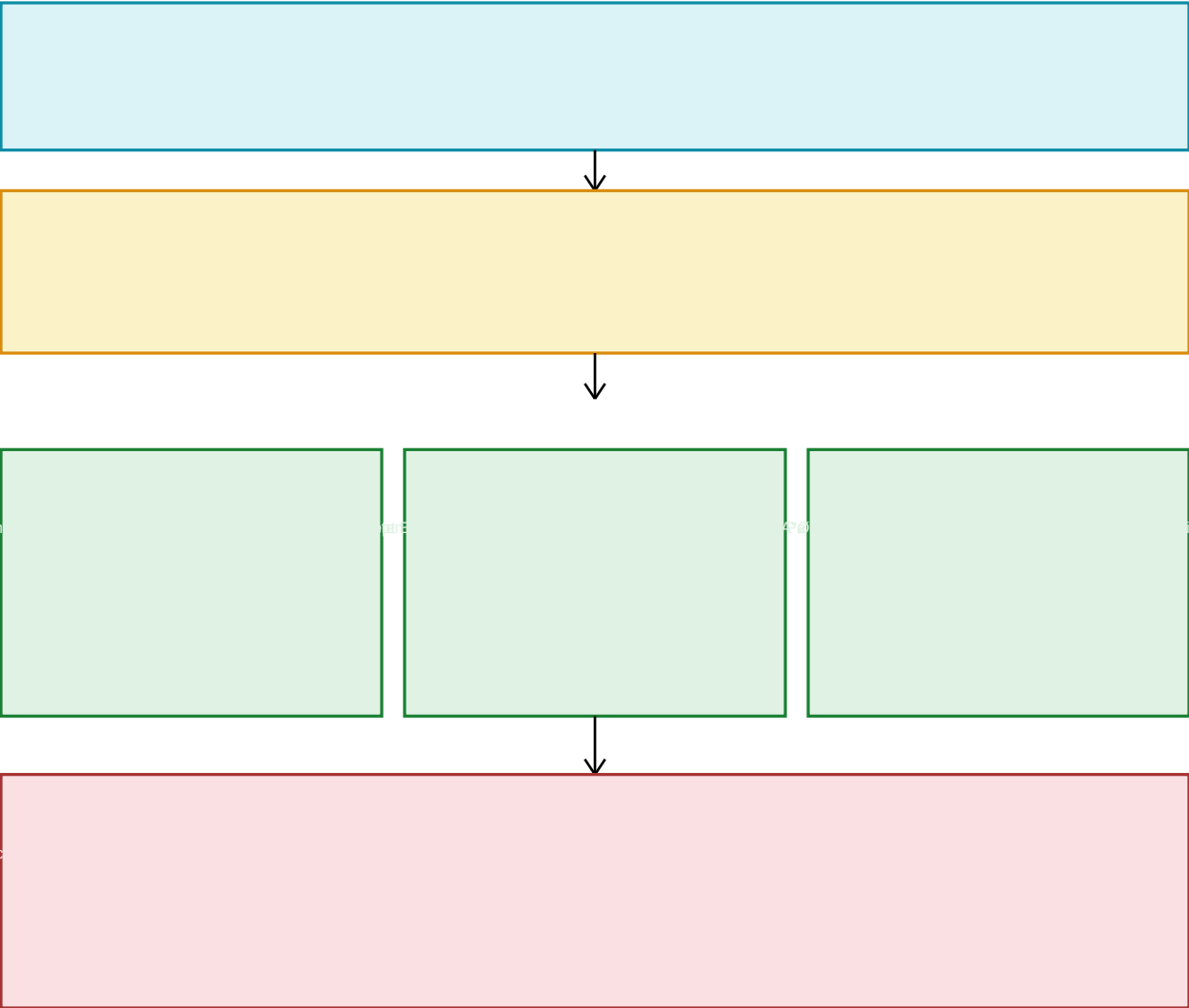
Team ID: SWTID-2026-3580

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Solution Architecture Diagram:

Completed architecture for the EduGenie web application, showing the request flow from the student interface through FastAPI and AI processing to the generated learning result.



Solution Architecture

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Component Description Table

Presentation Layer	Provides the student-facing interface for choosing a learning task, entering a problem statement, and displaying AI results.	HTML, CSS, JavaScript, Python
API Gateway	Receives browser requests, validates inputs, routes them to the correct module, and returns responses.	FastAPI, AWS Lambda
Auth / Configuration	The current MVP has no user-authentication system. API configuration is based on environment variables.	Python, AWS IAM, GitHub Actions
Core Logic Service	Coordinates Q&A, concept explanation, quiz generation, summarization, and feedback loops.	Python, Redis, Celery
External AI Integration	Generates learning content for Q&A, quizzes, summaries, and learning paths.	Google Gemini API, OpenAI
Local AI Model	Provides local concept explanation as described in the project design.	LaMini-Flan-T5-783M, Transformers, PyTorch
Architecture Notes	The MVP does not use a persistent database; application files and the local model are stored within the project/runtime environment.	Local project storage